SKILLNAVORA · AI PRODUCT MANAGEMENT & GENAI LAUNCHPAD

Machine Learning Products for PMs

Manage traditional ML products: problem types, data and labelling, metrics, working with data scientists, recommendation and personalisation, the model lifecycle and MLOps monitoring.

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How to use this guide

Each class has goals, concepts with Indian examples, key takeaways and a practice exercise, from basic to advanced. Finish with the final project to build something you can show in interviews.

	Class	Level
Class 1	**ML problem types**	Basic
Class 2	**Data and labelling**	Basic
Class 3	**Metrics and trade-offs**	Intermediate
Class 4	**Recommendations and personalisation**	Intermediate
Class 5	**ML use cases by industry and how they work**	Intermediate
Class 6	**Offline vs online evaluation and experimentation**	Intermediate
Class 7	**Model lifecycle and MLOps**	Advanced
Class 8	**Deployment, monitoring and drift**	Advanced
Class 9	**Writing an ML product requirements document**	Advanced
Class 10	**Case studies: fraud detection and recommendations end to end**	Advanced
Class 11	**Case studies: ETA prediction and churn prediction**	Advanced
Class 12	**Interview practice: ML product scenario questions**	Advanced
Class 13	**Working with data scientists and final project**	Advanced
	Toolkit: templates, cheat sheet, glossary, 15 interview questions, further reading	

BASIC · CLASS 1

ML problem types

Recognise which ML problem you're solving.

Type	Question	Example
Classification	Which category?	Fraud / not fraud, spam
Regression	How much?	Delivery time, price
Ranking / recommendation	What order?	Feed, search results, products
Clustering	What groups exist?	Customer segments
Forecasting	What next?	Demand, inventory
Anomaly detection	What's unusual?	Payment spikes, outages

Problem type	Question it answers	Product example	Typical output
Classification	Which category?	Is this transaction fraud? Is this review spam?	Label and probability
Regression	How much?	Delivery time, price estimate, demand	A number
Ranking / recommendation	What first?	Search results, home feed, 'frequently bought together'	Ordered list
Clustering	What groups exist?	Customer segments, similar stores	Group IDs
Anomaly detection	What is unusual?	Payment spike, server fault	Score and alert
Forecasting	What happens next?	Sales next week, rider demand by hour	Time series
NLP / generative	What does this say, or write this	Support triage, summaries, chat	Text
Computer vision	What is in this image?	Document OCR, product photo quality	Labels, boxes, text

Before using ML, ask: Could a rule or a simple lookup do 80% of the job? ML shines where rules are too many, change often, or depend on patterns humans cannot write down.

Key takeaways

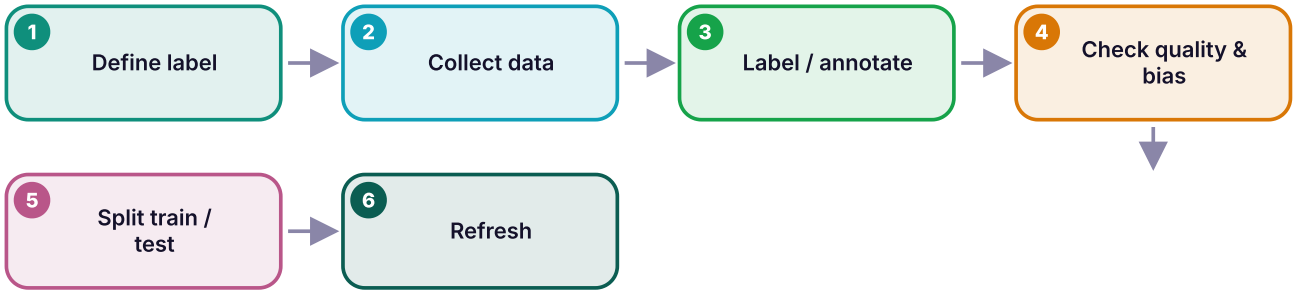
- Name the problem type first.
- Each type has different metrics.
- Many products combine types.

Practice exercise: Name the ML problem type for 8 features in apps you use.

BASIC · CLASS 2

Data and labelling

Get the data an ML product needs.



Data pipeline for ML

- Labels from outcomes (did the user default?), humans (annotators), or rules.
- Watch leakage (using future information), class imbalance (rare fraud), and bias.
- Labelling tools and vendors; clear guidelines and agreement checks.

Data quality checklist

Check	Question
Coverage	Do we have data for all user types, regions, devices and seasons?
Label quality	Do two annotators agree? What is the agreement rate?
Freshness	How old is it? Does behaviour change quickly?
Leakage	Does a feature contain information from the future or the answer itself?
Imbalance	Is the thing we want to predict rare (fraud 0.1%)?
Bias	Does the data under-represent groups?
Consent and privacy	Do we have the right to use it for this purpose?

Labelling approaches: use existing outcomes (a returned order = label), human annotators with guidelines, domain experts for hard cases, weak labels from rules, and active learning (label the examples the model is unsure about). Always keep a clean **golden set** for evaluation.

Key takeaways

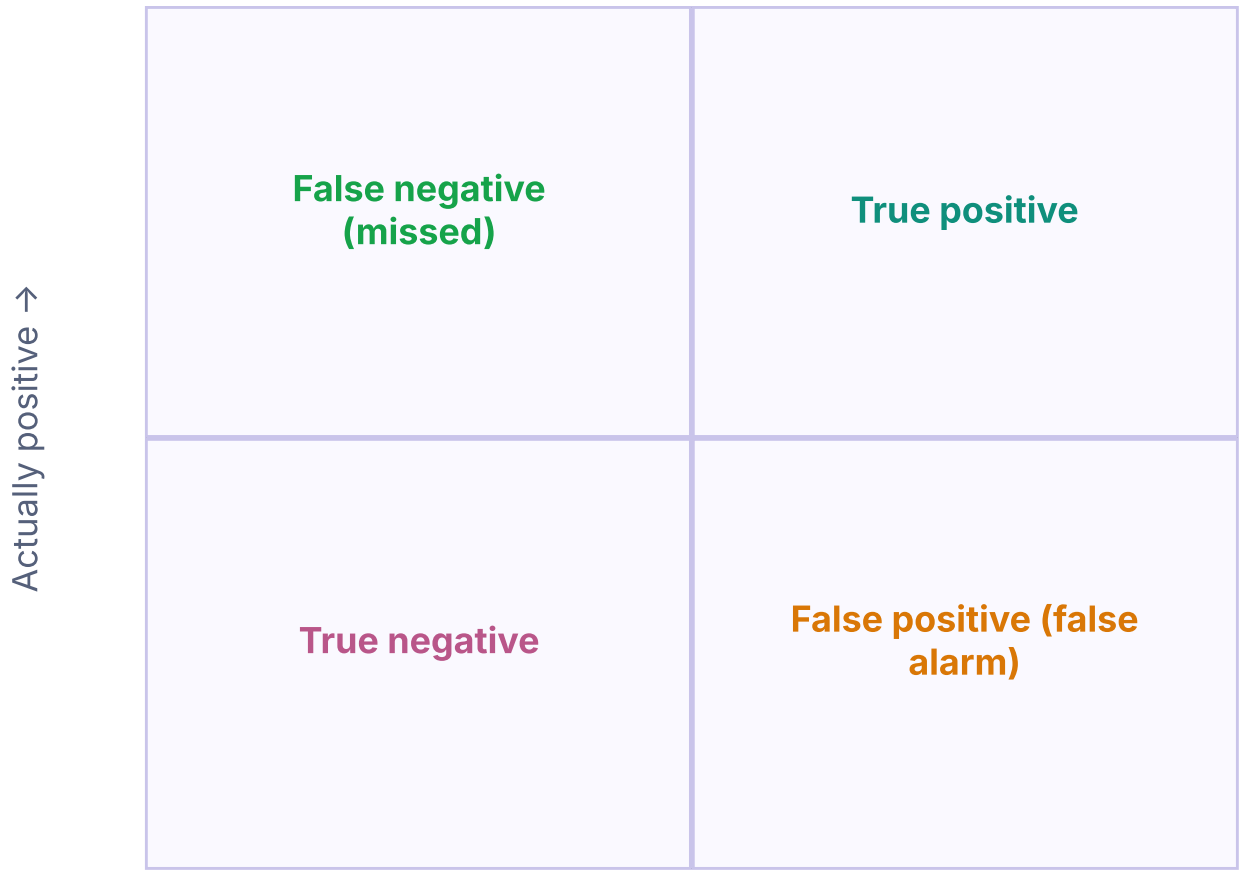
- Data quality decides model quality.
- Write labelling guidelines.
- Check leakage, imbalance and bias.

Practice exercise: Write labelling guidelines for classifying support tickets into 6 categories.

INTERMEDIATE · CLASS 3

Metrics and trade-offs

Choose model metrics that match business costs.



Confusion matrix (simplified)

- Precision = of flagged, how many were right; recall = of all positives, how many we caught.
- Choose thresholds by business cost: missing fraud vs annoying good customers.
- Link model metrics to product metrics (approval rate, losses, CSAT) with A/B tests.

Worked example: fraud detection with a cost matrix

Out of 100,000 payments, 200 are fraud. Model A flags 1,000 payments and catches 160 frauds (precision 16%, recall 80%). Model B flags 300 and catches 120 frauds (precision 40%, recall 60%).

	Model A	Model B
Frauds caught	160	120
Frauds missed (cost Rs 3,000 each)	40 × 3,000 = Rs 1,20,000	80 × 3,000 = Rs 2,40,000
Good users flagged (cost Rs 100 each in review and friction)	840 × 100 = Rs 84,000	180 × 100 = Rs 18,000
Total cost	Rs 2,04,000	Rs 2,58,000

Model A costs less overall, even though Model B has higher precision. Choose the threshold from **business cost**, not from a default of 0.5. Also watch the experience of the 840 good users flagged by A; add step-up verification instead of a hard block.

Metric	Meaning	When it matters
Precision	Of those flagged, how many were right	Cost of false alarms is high
Recall	Of all true cases, how many were found	Cost of misses is high
F1	Balance of the two	Single-number comparisons
AUC (ROC)	Ranking quality across thresholds	Comparing models
MAE / RMSE	Average error size (regression)	ETA, price, demand
NDCG, MRR, hit-rate	Quality of ranked lists	Search and recommendations

Key takeaways

- Precision vs recall is a business decision.
- Set thresholds by cost of errors.
- Validate with online experiments.

Practice exercise: For a loan-default model, decide which error is worse and how you'd set the threshold.

INTERMEDIATE · CLASS 4

Recommendations and personalisation

Understand how recommenders work and are measured.

- Approaches: popularity, collaborative filtering (people like you), content-based (similar items), and hybrid/deep models.
- Cold start for new users/items: onboarding questions, popular items, content features.
- Metrics: CTR, conversion, watch time, diversity, novelty, long-term retention.
- Guardrails: filter bubbles, fairness to sellers/creators, safety.

Recommendation approaches

Approach	How it works	Strength	Weakness
Popularity	Show what most people like	Simple, strong baseline	Not personal
Content-based	Recommend items similar to what you liked (using item features or embeddings)	Works for new items	Narrow, repetitive
Collaborative filtering	People like you liked these	Finds surprising matches	Cold start for new users and items
Hybrid / deep models	Combine signals with learned embeddings	Best accuracy	Complexity, cost, explainability
Bandits	Explore new options while exploiting known good ones	Learns fast	Needs careful design

- **Cold start:** use popularity, onboarding questions and item metadata until behaviour data exists.
- **Beyond accuracy:** diversity, novelty, fairness to sellers, freshness, and business rules (stock, margin).
- **Feedback loops:** showing only popular items makes them more popular; reserve slots for exploration.

Key takeaways

- Hybrid recommenders are common.
- Plan for cold start.
- Balance engagement with diversity and long-term value.

Practice exercise: Design recommendations for a regional-language news app.

INTERMEDIATE · CLASS 5

ML use cases by industry and how they work

See what ML does in e-commerce, fintech and services, with data and metrics.

Industry	Use case	How it works	Data needed	Metric
E-commerce	Search ranking	Rank products by relevance, price, stock and past clicks using a learning-to-rank model	Queries, clicks, add-to-cart, purchases	Search conversion, NDCG, zero-result rate
E-commerce	Recommendations	Suggest next products from behaviour and item embeddings	Views, orders, catalogue	CTR, add-to-cart, revenue per session
Quick commerce	ETA and demand prediction	Regression on distance, store load, rider supply, weather	Past orders and rides	ETA error, on-time rate, cost per order
Fintech	Credit scoring	Classification of default risk using bureau and behavioural data	Applications, repayment history	Approval rate, default rate, fairness gaps
Fintech	Fraud detection	Real-time scoring of transactions with rules plus models	Transactions, device, location	Fraud loss, false-positive rate, latency
Insurance	Claims triage	Classify claims and estimate severity; route to the right team	Past claims, documents	Cycle time, leakage
Healthcare	Triage and imaging support	Classify symptoms or images to prioritise cases	Labelled records and images	Sensitivity, specificity, clinician time saved
Marketing	Churn and next-best-action	Predict who will leave; choose the offer	Usage, billing, support history	Retention uplift, cost per saved customer

Pattern: prediction alone creates no value. Connect each prediction to a **decision and an action** (block, review, offer, reroute), and measure the action's outcome.

Key takeaways

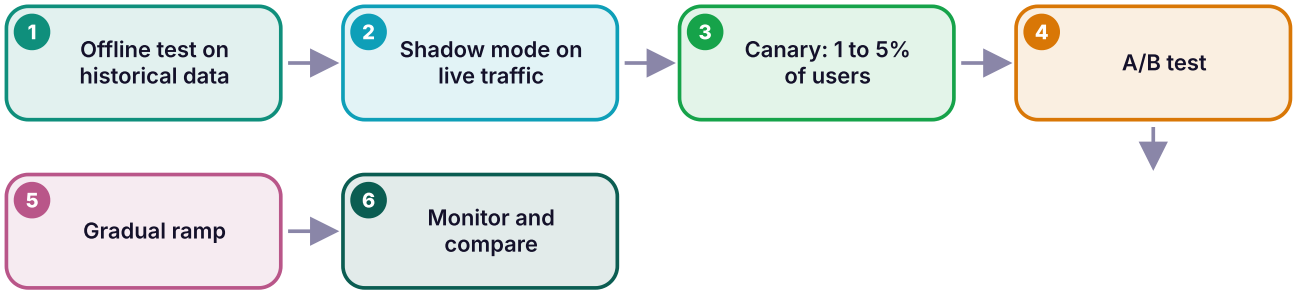
- A prediction needs a decision and an action to create value.
- Pick metrics that reflect the business outcome.
- Know the data each use case needs before committing.

Practice exercise: For one industry, write the prediction, the action it triggers, the data needed and the metric.

INTERMEDIATE · CLASS 6

Offline vs online evaluation and experimentation

Test models safely before and after launch.



Staged evaluation of an ML model

- **Offline evaluation** (precision, AUC, NDCG) is fast and cheap but can mislead: history is not the future, and the model changes user behaviour.
- **Shadow mode:** the model scores real traffic silently so you compare it to the current system without affecting users.
- **Online A/B test:** randomise users; primary metric tied to the business (conversion, fraud loss), guardrails (latency, complaints, fairness), enough sample and time to cover weekly cycles.
- **Interleaving** (for ranking) mixes results from two rankers in one list to detect preference with less traffic.
- **Counterfactual and bandit methods** help when random tests are costly.
- **Novelty effects:** early excitement can fade; run long enough.

Pitfall: a model with a better offline score can lose online because of latency, odd edge cases, or feedback loops. Always confirm online.

Key takeaways

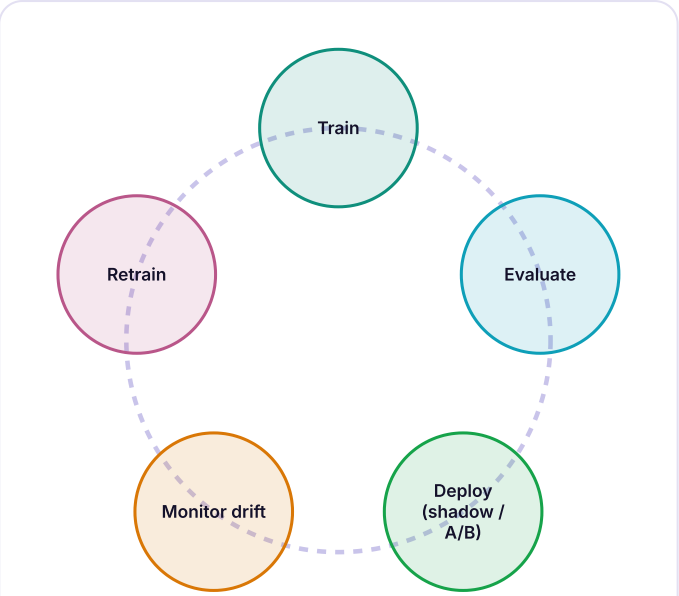
- Offline metrics are necessary, not sufficient.
- Use shadow, canary and A/B stages.
- Pick guardrails as carefully as the primary metric.

Practice exercise: Design a staged evaluation plan for a new search ranking model.

ADVANCED · CLASS 7

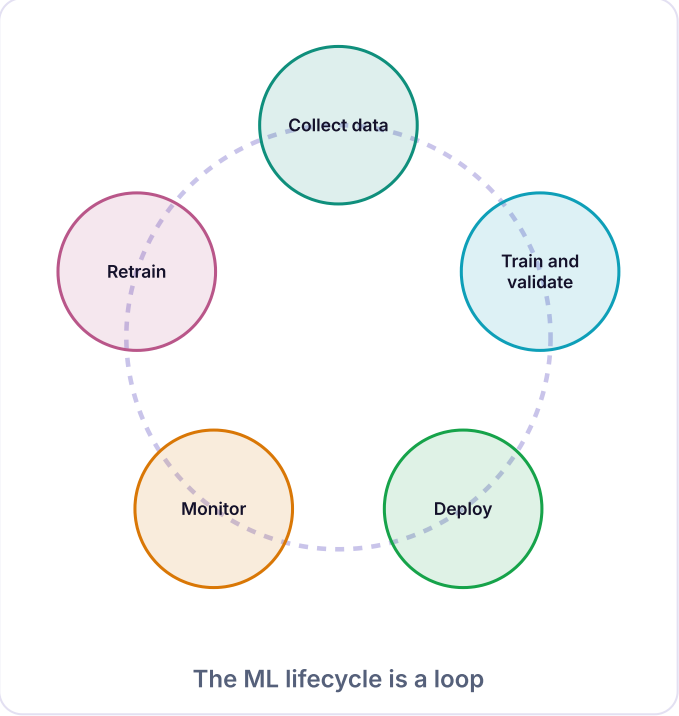
Model lifecycle and MLOps

Keep models working after launch.



ML lifecycle

- Shadow mode and gradual rollout before full launch.
- Monitor data drift, prediction drift and business metrics; alert on drops.
- Model registry and versioning; rollback plans.
- Model cards documenting purpose, data, limits.



The ML lifecycle is a loop

MLOps is the practice of running that loop reliably: version data, code and models; automate training and testing; deploy safely (shadow, canary); monitor quality and cost; retrain on triggers.

Key takeaways

- Launch is the start of the lifecycle.
- Monitor drift and business impact.
- Always have a rollback.

Practice exercise: Write a monitoring plan for a delivery-ETA model.

ADVANCED · CLASS 8

Deployment, monitoring and drift

Keep ML models healthy in production.

Topic	What PMs should know
Batch vs real-time	Batch scores many items on a schedule (cheap, stale). Real-time scores on request (fresh, needs low latency and robust infrastructure).
Latency budget	Total time allowed (for example 100 ms for a payment decision), split across feature lookup, model and rules.
Feature store	A shared place for computed features so training and serving use identical logic (avoids training-serving skew).
Model registry and versioning	Every model has a version, data lineage, metrics, owner, and rollback path.
Data drift	Inputs change (new device types, new city). Detect with distribution checks.
Concept drift	The relationship changes (new fraud tactics). Detect with quality metrics against fresh labels.
Retraining triggers	Schedule, drift alert, performance drop, major product change.
Fallbacks	Rules, previous model or a safe default when the model fails or is slow.
Cost	Compute for training and serving, storage, labelling; track cost per prediction.

Monitoring dashboard: data quality, input drift, prediction distribution, business KPIs, latency, errors, cost, fairness slices, and alerts with owners.

Key takeaways

- Plan monitoring and fallbacks before launch.
- Drift is normal; build retraining into the roadmap.
- Track cost and latency, not only accuracy.

Practice exercise: Design a monitoring and fallback plan for a real-time fraud model.

ADVANCED · CLASS 9

Writing an ML product requirements document

Specify ML features so data scientists and engineers can build the right thing.

Section	What to write
Problem and users	Who benefits, what decision changes, current process and baseline
Success metrics	Business metric, model metric, guardrails, targets and the acceptable range
Data	Sources, volume, labels, freshness, consent, known gaps
Model requirements	Latency, throughput, explainability, fairness, privacy, offline vs real-time
Thresholds and actions	What happens at each score band (approve, review, block)
Fallback and human review	What happens when the model is unsure or down
Evaluation plan	Offline set, shadow, A/B, timeline
Risks	Bias, drift, abuse, regulation, dependence on a vendor
Milestones	Data readiness, baseline, prototype, pilot, rollout, with uncertainty acknowledged

Estimating ML work: data work often takes longer than modelling. Plan an early **baseline** (rules or a simple model) to learn what is possible, then set targets with ranges, not promises.

Key takeaways

- Specify decisions, thresholds and fallbacks, not just 'build a model'.
- Start with a baseline.
- Express timelines as ranges.

Practice exercise: Write an ML PRD outline for a churn-prediction feature.

ADVANCED · CLASS 10

Case studies: fraud detection and recommendations end to end

Walk two complete ML products from problem to monitoring.

Case 1: real-time payment fraud (fintech)

1. **Problem:** fraud losses are 0.12% of volume; manual review is slow.
2. **Baseline:** rules (new device + high amount) catch 45% of fraud with many false alarms.
3. **Model:** gradient-boosted trees on device, velocity, location, payee and history; score under 50 ms.
4. **Actions:** low risk approve; medium step-up (OTP or biometric); high block and review.
5. **Evaluation:** offline on past fraud; shadow for 2 weeks; A/B on 10%.
6. **Guardrails:** payment success rate, step-up completion, complaints from blocked genuine users.
7. **Monitoring:** daily drift, weekly new-pattern review, retrain monthly.

Case 2: home-feed recommendations (e-commerce)

1. **Problem:** low repeat rate; users scroll but do not find items.
2. **Baseline:** popular items by category.
3. **Model:** two-stage: candidate generation (embeddings from views and orders) then ranking (clicks and purchase probability with price, stock, margin).
4. **Cold start:** onboarding choices and trending items for new users.
5. **Metrics:** add-to-cart per session, orders per user, diversity, seller exposure; guardrail: page speed.
6. **Risks:** popularity bias, filter bubbles, stale stock; fix with exploration slots and real-time inventory checks.

Key takeaways

- Start with a rules or popularity baseline.
- Connect scores to actions with thresholds.
- Plan guardrails and monitoring from day one.

Practice exercise: Choose one case and write the full plan using the seven steps.

ADVANCED · CLASS 11

Case studies: ETA prediction and churn prediction

Work through two ML products end to end (illustrative).

Case 1: ETA prediction for 10-minute delivery

1. **Problem:** customers cancel when ETAs are wrong; the rule-based ETA is off by 4 minutes on average.
2. **Data:** distance, store load, picker queue, rider supply, traffic, weather, time of day.
3. **Model:** regression (gradient boosting) predicting delivery time and a prediction interval.
4. **Metric:** mean absolute error (MAE) and share of orders within 2 minutes of the promised ETA; business: cancellation rate and NPS.
5. **Action:** show ETA ranges, adjust dispatch when late risk is high.
6. **Risks:** underestimates hurt trust more than overestimates; add a safety buffer by percentile (p80).
7. **Monitoring:** weekly error by city and hour; drift alert on rainy days.

Case 2: Churn prediction for a subscription app

1. **Problem:** 8% monthly churn; retention offers are expensive.
2. **Target:** will cancel in the next 30 days.
3. **Evaluation:** precision at top 10% (who to contact), uplift (did the offer change behaviour?).
4. **Action:** send tailored offers only to users who are likely to churn and likely to respond; leave others alone.
5. **Experiment:** randomise a hold-out group to measure true saves.
6. **Risk:** offering discounts to people who would have stayed.

Key takeaways

- Match metric to the decision.
- Use uplift thinking when actions cost money.
- Monitor by segment and condition.

Practice exercise: Write a plan for another prediction product (stock-out prediction) using the same steps.

ADVANCED · CLASS 12

Interview practice: ML product scenario questions

Practise ML product thinking without code.

Q. How would you build a fraud detection product?
Model answer: I would define loss and friction costs, start with rules as a baseline, collect labelled fraud outcomes, build a model scoring in real time, set thresholds from a cost matrix with step-up verification for medium risk, run shadow mode then A/B, and monitor drift, false positives and complaints. I would work with data science on features and with risk on policy.
Why it works: It covers baseline, labels, thresholds, rollout and monitoring.

Q. The model's accuracy is 99% but the business is not happy. Why?
Model answer: Accuracy can hide class imbalance: if fraud is 1% of cases, predicting 'no fraud' gives 99%. I would use precision, recall, cost-based metrics and business outcomes by segment.
Why it works: Shows understanding of imbalance.

Q. How do you decide when to retrain a model?
Model answer: Triggers: scheduled cadence, data or concept drift, performance drop on fresh labels, or product changes. I would define monitoring thresholds, an automated pipeline and a rollback plan.
Why it works: Operational thinking.

Q. How do you explain a model decision to a customer?
Model answer: Use reason codes tied to understandable factors, avoid revealing sensitive internals that enable gaming, provide an appeal path, and keep logs for audits.
Why it works: Balances transparency and safety.

Key takeaways

- Use cost-based thinking.
- Know accuracy pitfalls.
- Plan retraining and monitoring.

Practice exercise: Answer the fraud question using a cost matrix with your own numbers.

ADVANCED · CLASS 13

Working with data scientists and final project

Collaborate well and ship an ML feature.

- PM brings: problem, users, business costs of errors, constraints, success metric.
- DS brings: feasibility, data needs, model choice, offline metrics.
- Agree early: baseline (simple rules), target metric, timeline for experiments.

Final project

Write an ML PRD for a fraud or recommendation feature: problem type, labels and data, metrics and threshold logic, A/B plan, monitoring, model card.

Key takeaways

- Start from a simple baseline.
- Agree metrics and costs of errors early.
- Document with a model card.

Practice exercise: Complete the ML PRD.

TOOLKIT

Templates

ML PRD extras

Field	Fill in
Problem type	
Label definition & source	
Baseline	
Offline metric & target	
Threshold logic (cost of errors)	
Online A/B metric	
Monitoring & drift alerts	
Rollback plan	
Model card link	

TOOLKIT

Cheat sheet

Term / tool	Meaning / use
Precision	TP ÷ (TP + FP)
Recall	TP ÷ (TP + FN)
Threshold	Score cut-off for a decision
Drift	Data or behaviour changing over time
Shadow mode	Model runs without affecting users
Cold start	No history for new users/items
Precision / recall	Right when flagged / found among all true
Threshold choice	Use business cost, not 0.5
Stages	Offline, shadow, canary, A/B, ramp
Drift	Inputs change vs relationships change

Glossary

Label	The correct answer used for training
Leakage	Training on information unavailable at prediction time
Baseline	A simple method used for comparison
Golden set	A trusted labelled set for evaluation
Threshold	The score above which an action is taken
Cost matrix	Money cost of each type of error
Shadow mode	Scoring live traffic without acting on it
Canary	Release to a small share of users first
Training-serving skew	Difference between training and live feature calculation
Cold start	No history for a new user or item
NDCG	Ranking quality metric
Feature	An input signal to a model
Overfitting	Model memorises training data and fails on new data
Active learning	Label the examples the model is least sure of

INTERVIEW PREP

15 interview questions with answers

Q1. How is ML PM different from regular PM?
Probabilistic outputs, data dependency, metric trade-offs, experimentation and ongoing monitoring/retraining.

Q2. Precision vs recall trade-off?
Choose based on the cost of false positives vs false negatives; set thresholds accordingly and validate online.

Q3. How do you handle cold start?
Onboarding preferences, popular/trending items, content-based features, and exploration.

Q4. What is drift?
Change in input data or relationships over time that degrades model quality; detect via monitoring and retrain.

Q5. What goes in an ML PRD?
Problem type, labels/data, metrics and thresholds, baseline, experiment plan, monitoring, rollback, model card.

Q6. How do you decide whether ML is the right solution?
Check if rules or a lookup solve most of the problem, whether there is enough quality data and labels, whether wrong predictions are tolerable, and whether value is measurable. Otherwise start without ML.

Q7. Precision or recall: how do you choose?
From the cost of each error type. If misses are costly (fraud), favour recall with review steps; if false alarms are costly (blocking customers), favour precision. Pick the threshold with a cost matrix.

Q8. What is data leakage?
Using information in training that would not be available at prediction time, which inflates offline results and fails in production.

Q9. What is the cold-start problem?
Not having enough data for a new user or item. Solutions include popularity, metadata, onboarding questions and exploration.

Q10. Offline metrics improved but the A/B test did not. Why?
Offline data may not reflect new behaviour, there may be latency or edge-case problems, the metric may not match the business goal, or feedback loops and novelty effects are in play.

Q11. What is model drift?
Data drift is a change in inputs; concept drift is a change in the relationship between inputs and outcomes. Both reduce performance and need monitoring and retraining.

Q12. How do you work with data scientists?
Share the business problem and metric, agree baselines and thresholds, review error analysis together, plan data work early, and make trade-offs explicit.

Q13. How would you measure a recommendation system?
Offline NDCG and hit rate, online click-through, add-to-cart, conversion and revenue per user, plus diversity, fairness to sellers and long-term retention as guardrails.

Q14. What is a feature store?

A shared system that computes and serves features consistently for training and prediction, avoiding training-serving skew.

Q15. How would you handle a model failure in production?

Use a fallback (rules or previous model), alert owners, roll back, analyse root cause, and add tests and monitoring to prevent recurrence.

LEARN MORE

Further reading and tools

- Google: Rules of ML ↗
- Google ML Crash Course ↗
- Model cards paper (Google) ↗
- SkillNavora Statistics & ML track ↗
- Google Rules of ML ↗
- Google ML Crash Course ↗
- Made With ML (MLOps) ↗

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